

Mikael GALLEGO

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SUMMARY

Second-year Master's physics student specializing in light-matter interactions at Université de Bordeaux. Currently at LOMA, analyzing EEG signals during sleep under the internship of Pr. Françoise ARGOUL. Seeking Ph.D. positions beginning in Autumn 2026. Particularly interested in research with applications in human health or environmental conservation. Born 23 May 2003 in Denver, CO, USA.

EDUCATION

Université de Bordeaux | *Master Physique fondamentale et applications* 2024 — Now

Specialization: Light, Matter, and iNteractions (LMN)

Awarded France Excellence Graduate Scholarship by Campus France USA (M1, M2)

Awarded Excellence Fellowship by the LIGHTS&T Graduate Program (M1)

North Carolina State University | *B.S. Physics, B.S. Mathematics* 2020 — 2024

Graduated Magna Cum Laude

Earned Dean's List in 5 semesters (GPA of 3.5 / 4.0 or higher)

Completed French Certificate

COURSEWORK

- Quantum Mechanics, Quantum Optics
- Laser Physics and Nonlinear Optics
- Electrodynamics
- Statistical Mechanics
- Optics of Nanomaterials
- Nanophysics, Nanophotonics
- Numerical Analysis, Physics Simulations
- Real and Complex Analysis, Group Theory

SKILLS

- **Computer languages and software:** C, Python, Wolfram, L^AT_EX, Microsoft Office, Linux/UNIX, OSLO, MPI, IBM Spectrum LSF, LabView.
- **Competencies:** High performance/parallel/cluster computing, laser safety certified, cleanroom and precision machining experience, team leading, scientific writing and presentation, data analysis and presentation.
- **Human languages:** English (native), French (B2).

EXPERIENCE

Laboratoire Ondes et Matière d'Aquitaine (LOMA) | *Intern* Feb. 2026 — Now

- Organize a comprehensive literature and mathematical review of time-frequency analysis in neuroscience.
- Analyze time-series EEG data from the open ANPHY-Sleep database. Implement techniques like the continuous wavelet transform to detect and characterize sleep spindles in the N2 sleep stage.

Integration from Material to System Laboratory (IMS) | *Intern* Jan. 2025 — July 2025

- Designed an pump-probe setup to be used with a THz antenna array for nondestructive testing, offering scan time speedups of several orders of magnitude over traditional systems. Aligned optical elements. Used polarizers and beam splitters to fine-tune pump and probe beam powers.
- Modeled systems of parabolic and ellipsoidal mirrors in OSLO for beam focusing. Designed and machined a stand for a galvanometric scanner. Used software to control the infrared laser source and ensured its stability over time. Defended results in the M1 thesis and corresponding oral defense.

Nonlinear Artificial Intelligence Lab (NAIL) | *Research Assistant* Feb. 2023 — Feb. 2026

- Investigated the effect of spatial disorder 'taming' chaotic systems. Coded an array of damped, driven, coupled pendulums in C and parallelized this system with MPI for runs on a high-performance computing cluster. Formatted batch jobs in Linux using IBM Spectrum LSF. Analyzed data and visualized results in Mathematica. Presented at the NCSU Undergraduate Research Symposium.

- Introduced disorder to the array via pendulum-length perturbations to drive chaotic behavior into a periodic regime. Mapped ordered and chaotic regimes in parameter space to show chaos emerging from large seas of periodic evolution, determined by maximal Lyapunov exponent approximation. Provided a complete dynamical picture by examining up to (and including) the fused-array infinite-coupling limit and zero-pendulum-length maximum-disorder limit. Showed results for constant and variable mean pendulum length. Published in Phys. Lett. A.

Single Photon Quantum Mechanics (Senior Lab) | *Student*

Aug. 2023 — Dec. 2023

- Conducted experiments in a team of two students to investigate the quantum mechanical nature of photons. Measured coincidence counts for entangled photon pairs and photon-photon interference. Conducted Bell tests. Collected and analyzed data with LabView. Troubleshot experiments and optimized lab processes. Prepared detailed laboratory results, presented to faculty of the NCSU Department of Physics.
- Designed optical systems for generating and measuring correlated single photons. Aligned precision optical components to split a photon pump beam into two cones of correlated photons via BBO crystal. Used bandpass filters and collimators to direct broadband photons into detectors. Worked in a darkroom.

NCSU Physics Department | *Employee*

April 2022 — July 2024

- Constructed classroom demonstrations for physics and engineering professors. Troubleshot circuits and mechanical systems to fit professors' needs. Provided assistance in the campus' precision machining shop.
- Hosted tables for the NC Museum of Natural Science's "Astronomy Days", which attracts thousands of people. Organized the annual LEAP workshop for high school students, displayed and explained physics demonstrations.

PUBLICATION

- **M. Gallego**, J. F. Lindner, A. Radhakrishnan, W. L. Ditto, Lyapunov landscape elucidates disorder-taming-chaos in arrays of coupled pendula, Physics Letters A 579 (2026) 131483.
DOI: 10.1016/j.physleta.2026.131483

CONFERENCES

- Presented: "*Photoconductive terahertz antenna source array for nondestructive testing*", EUR LIGHT Master Workshop, Université de Bordeaux, June 2025.
- Presented: "*Disorder Taming Chaos in Coupled Pendulum Arrays*", Undergraduate Research Symposium, North Carolina State University, April 2024.
- Presented: "*Single Photon Quantum Mechanics*", Senior Physics Lab Presentations, North Carolina State University, December 2023.